

Week 4: Work and Kinetic Energy – Tutorial Sheet

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Use the Master Procedure: identify, mark the two states, draw the free-body diagram, find the work of each force with its sign, predict, select the method, execute $W_{\text{tot}} = \Delta K$, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A sled on level ground

A 15.0 kg sled is pulled 10.0 m along level ground by a rope with tension 60 N, 20° above the horizontal. The coefficient of kinetic friction is 0.10. Find the work done by each force, the total work, and the final speed if it starts from rest.

Solution. Vertical: $n = mg - T \sin 20^\circ = 147 - 20.5 = 126.5 \text{ N}$, so $f_k = 0.10(126.5) = 12.65 \text{ N}$. $W_T = Ts \cos 20^\circ = (60)(10.0)(0.940) = +564 \text{ J}$; $W_f = -(12.65)(10.0) = -127 \text{ J}$; $W_w = W_n = 0$. Total: 437 J. Check (Method II): $(60 \cos 20^\circ - 12.65)(10.0) = (56.4 - 12.65)(10.0) = 437 \text{ J}$. Work-energy: $437 = \frac{1}{2}(15.0)v^2 \Rightarrow v = 7.6 \text{ m/s}$.

A2. A spring-launched ball

A 0.300 kg ball is pushed against a horizontal spring ($k = 500 \text{ N/m}$), compressing it 0.20 m, and released on a smooth surface. Find its speed when it leaves the spring.

Solution. Work done by the spring: $\frac{1}{2}kx^2 = \frac{1}{2}(500)(0.20)^2 = 10.0 \text{ J}$. This is the total work (no friction), so $\frac{1}{2}mv^2 = 10.0$ and $v = \sqrt{2(10.0)/0.300} = 8.2 \text{ m/s}$.

B. Practice

B1. A ramp with friction

A 10.0 kg crate slides 4.0 m down a 30° ramp with $\mu_k = 0.25$, starting from rest. Find the work done by gravity, by friction and by the normal force, and the final speed.

B2. Stopping a truck

A 8000 kg truck at 20 m/s is stopped by a constant braking force of 32 kN. Find the stopping distance. How far would it need at 40 m/s?

B3. A varying force

The force on a particle is $F_x = 10x - 2x^2$ (SI units). Find the work done from $x = 0$ to $x = 3.0 \text{ m}$.

B4. A spring and friction

A 1.0 kg block is pressed against a spring ($k = 200 \text{ N/m}$) compressed 0.30 m and released on a rough floor ($\mu_k = 0.30$). Find its speed as it leaves the spring and the total distance it slides.

B5. Power of a lift

A motor raises an 800 kg load at a constant 2.0 m/s. What is the power developed (neglecting friction)? How much energy does it supply in one minute?

B6. Challenge: the method is not given

A 60 kg skier slides down a 25 m long slope inclined 20° to the horizontal with $\mu_k = 0.10$, and then continues along a level patch with $\mu_k = 0.20$ until she stops. How far does she slide on the level patch?

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. A rope inclined at 35° above the horizontal pulls a crate 8.0 m along a level floor with a constant tension of 60 N. The work done by the tension is

- (A) 49.1 J (B) 275 J (C) 393 J (D) 480 J (E) 786 J

E2. A force $F_x = 6x^2 - 4x$ (F in N, x in m) acts on a particle that moves from $x = 0$ to $x = 2.0$ m. The work done by the force is

- (A) 4.00 J (B) 8.00 J (C) 16.0 J (D) 20.0 J (E) 32.0 J

E3. The net work needed to increase the speed of a 1200 kg car from 15.0 m/s to 25.0 m/s is

- (A) $1.20 \times 10^4 \text{ J}$
(B) $6.00 \times 10^4 \text{ J}$
(C) $1.35 \times 10^5 \text{ J}$
(D) $2.40 \times 10^5 \text{ J}$
(E) $3.75 \times 10^5 \text{ J}$

Items E4 and E5 share this information: A 900 kg car travelling at 20.0 m/s is brought to rest by a constant braking force over a distance of 40.0 m.

E4. The magnitude of the braking force is

- (A) 450 N (B) 2250 N (C) 4500 N (D) 8820 N (E) 9000 N

E5. The same car with the same braking force starts at twice the speed. The stopping distance is

- (A) 20 m (B) 80 m (C) 120 m (D) 160 m (E) 320 m

E6. A spring of force constant 400 N/m is slowly stretched from an extension of 0.050 m to 0.150 m . The work done by the stretching force is

- (A) 4.0 J (B) 4.5 J (C) 8.0 J (D) 9.0 J (E) 40.0 J

E7. Which of the following statements is/are true?

- I. A force that is always perpendicular to the displacement does no work.
- II. The total work done by the centripetal force during one full revolution of uniform circular motion is zero.
- III. Friction can never do positive work.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

Items E8 and E9 share this information: Blocks of mass 3.0 kg and 5.0 kg hang from the ends of a light rope over a frictionless, massless pulley and are released from rest. The heavier block descends 2.0 m .

E8. The work done by the rope tension on the 3.0 kg block (which rises) is

- (A) -73.5 J (B) 0.0 J (C) 58.8 J (D) 73.5 J (E) 98.0 J

E9. The net work done on the 5.0 kg block as it descends is

- (A) -24.5 J (B) 49.0 J (C) 98.0 J (D) 171.5 J (E) 24.5 J

E10. A 4.0 g bullet travelling at 300 m/s stops after penetrating 12 cm into a wooden block. The average stopping force is

- (A) 1.50 N
(B) 15.0 N
(C) 1500 N
(D) $1.50 \times 10^5 \text{ N}$
(E) $1.50 \times 10^6 \text{ N}$

Answers

B1 $W_g = +196 \text{ J}$, $W_f = -85 \text{ J}$, $W_n = 0$; $v = 4.7 \text{ m/s}$

B2 50 m ; 200 m (four times)

B3 27 J

B4 $v = 4.0 \text{ m/s}$; $d = 3.1 \text{ m}$

B5 $P = mgv = 15.7 \text{ kW}$ ($\approx 21 \text{ hp}$); $9.4 \times 10^5 \text{ J}$

B6 kinetic energy at the bottom $3.6 \times 10^3 \text{ J}$ ($v = 11 \text{ m/s}$); distance on the level patch 31 m

Exam-style practice: answer key and brief solutions

Key: E1 C E2 B E3 D E4 C E5 D E6 A E7 B E8 D E9 E E10 C

E1 (C). $W = Fs \cos \phi$ with $\phi = 35^\circ$, the angle between the force and the displacement: $W = (60)(8.0) \cos 35^\circ = 393 \text{ J}$. Only the component along the motion counts, so $W < Fs = 480 \text{ J}$.

E2 (B). $W = \int_0^{2.0} (6x^2 - 4x) dx = [2x^3 - 2x^2]_0^{2.0} = 16 - 8 = 8.0 \text{ J}$. The final force $F(2.0) = 16 \text{ N}$ times the distance would give 32 J , but the force is not constant: it is zero at $x = 0.67 \text{ m}$ and negative before that, so integrate.

E3 (D). $W_{\text{net}} = \Delta K = \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}(1200)(625 - 225) = 2.40 \times 10^5 \text{ J}$. The squares are subtracted, not the speeds: using $(v_2 - v_1)^2$ gives $6.0 \times 10^4 \text{ J}$.

E4 (C). Work-energy: $-Fd = 0 - \frac{1}{2}mv^2$, so $F = \frac{mv^2}{2d} = \frac{(900)(400)}{2(40.0)} = 4500 \text{ N}$. The braking force does negative work, so $W = -Fd = -1.8 \times 10^5 \text{ J} = \Delta K$.

E5 (D). $d = \frac{mv^2}{2F} \propto v^2$. Doubling v multiplies the distance by 4: $d = 160 \text{ m}$. The work needed is four times larger because $K \propto v^2$.

E6 (A). $W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}(400)(0.0225 - 0.0025) = 4.0 \text{ J}$. The squares are subtracted: the work from the natural length to 0.150 m alone is 4.5 J , which includes the first 0.050 m .

E7 (B). I is true: $W = Fs \cos 90^\circ = 0$. II is true: the centripetal force is always perpendicular to the velocity, so the speed and K do not change. III is false: static friction does positive work on a box carried by an accelerating lorry or conveyor belt. Answer: I and II only.

E8 (D). For the pair $a = \frac{m_2 - m_1}{m_1 + m_2}g = 2.45 \text{ m/s}^2$ and the tension is $T = \frac{2m_1m_2}{m_1 + m_2}g = 36.75 \text{ N}$. The tension is upward, along the displacement of the rising block: $W_T = Ts = (36.75)(2.0) = 73.5 \text{ J}$, positive. (On the descending block the same tension does -73.5 J , so the rope does no net work.)

E9 (E). Net work $= W_w + W_T = m_2gh - Ts = (5.0)(9.80)(2.0) - (36.75)(2.0) = 98.0 - 73.5 = 24.5 \text{ J}$, which equals its gain in kinetic energy $\frac{1}{2}m_2v^2$ with $v^2 = 2as = 9.80$. Gravity alone does 98.0 J , but the tension takes some of it away.

E10 (C). Convert before substituting: $m = 4.0 \times 10^{-3} \text{ kg}$, $d = 0.12 \text{ m}$. Then $F = \frac{mv^2}{2d} = \frac{(4.0 \times 10^{-3})(300)^2}{2(0.12)} = 1500 \text{ N}$. Leaving d in cm gives 30 N , and leaving m in grams gives $1.5 \times 10^6 \text{ N}$: check the magnitude against a bullet that stops in a few centimetres.